

UGTS-KC 3.6.3 – SARA

Seed-Address Referential Algebra

Public deterministic-wallet standards packed into the referential substrate

Integrated mechanisms

BIP39 entropy, checksum and 11-bit word cells; an ordered mnemonic cell complex; NFKD and the standard seed transition; a typed BIP32 derivation tree with distinct normal and hardened edges; the BIP84 P2WPKH route; compressed public-key and HASH160 projection; witness-program/script relations; Bech32 and Bech32m validation; public-only certificates; an explicit authorization guard; non-enumerating search-space metrics; and a canonical handoff to the UGTS event calculus.

Non-negotiable boundary

This version does not implement or document a real-wallet brute-force engine. It contains no mnemonic/passphrase enumeration, private-key scan, target-address preimage loop, signing, transaction broadcast, balance targeting or funds-transfer path. The supplied address is used only as a public checksum, witness-program and script fixture.

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Version 3.6.3 – 18 August 2026

Standards-derived cryptographic operators are separated from engineering-derived UGTS integration and rejected attack scopes.

Abstract

Version 3.6.3 extends the 3.6.2 packed geometric-topological substrate with a cryptographic representation profile. The new profile does not turn a wallet into geometry and does not claim that topology weakens cryptography. It uses the same referential discipline: every object has a type, every transform has a domain and codomain, every packed record has a schema, every guard has an explicit reason code, and every public result has lineage.

The standards path is represented as

entropy → checksum → 11-bit word cells → mnemonic
 → transient seed → HD tree → public key
 → witness program → address.

The implementation validates official public test vectors, decodes public addresses, produces public-only certificates, validates self-owned known inputs, and calculates optimistic feasibility metrics without generating candidates. It contains 45 new atomic operators, 19 content-addressed definitions, 20 claims-ledger entries, 42 new tests and all 153 retained tests from earlier versions, for 195 passing tests in total.

FORMAL DECISION

The requested “brute-force pentesting” concept is admitted only as a bounded, non-operational feasibility analysis and authorization gate. A real third-party mainnet key search is rejected. The safe technical core is standards conformance, public decoding, self-owned deterministic reconstruction, watch-only consistency, test-network experimentation and search-space measurement.

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1 Version lineage, source classes and scope

1.1 Inherited substrate discipline

UGTS-KC 3.6.2 made the authoritative object a packed, typed relation substrate rather than a renderer. SARA preserves that architecture. Cryptographic objects are not collapsed into one bit or one address: entropy, word indices, mnemonic text, seed, chain code, private scalar, public key, script, address, authorization and lineage are distinct types.

The update uses three visibly separate evidence classes:

Class	Contains	Examples
Standards-derived	Transformations explicitly specified by Bitcoin Improvement Proposals.	BIP39 checksum and seed transition; BIP32 child derivation; BIP84 path; BIP141 witness program; BIP173/BIP350 checksums.
Engineering-derived	Typed integration that preserves the standards while adding bounds, references and reason codes.	Word-cell complex, authorization guard, public-only certificate, UGTS handoff, schema and content-addressed definitions.
Rejected/unsupported	Capabilities that are not needed for conformance and would cross the authorization boundary.	Third-party secret search, passphrase spraying, private-key scanning, signing, broadcasting, balance targeting and funds transfer.

1.2 Security decision

Public visibility of an address does not grant permission to recover or use its spending key. The package therefore treats

bc1q7ydrtdn8z62xhslqyqytyt38mm4e2c4h3mxjkug

as a public syntax fixture only. The requester-provided ownership/context label is recorded as unverified external metadata and never used as legal or cryptographic proof.

UGTS-KC 3.6.3 SARA: public deterministic wallet pipeline

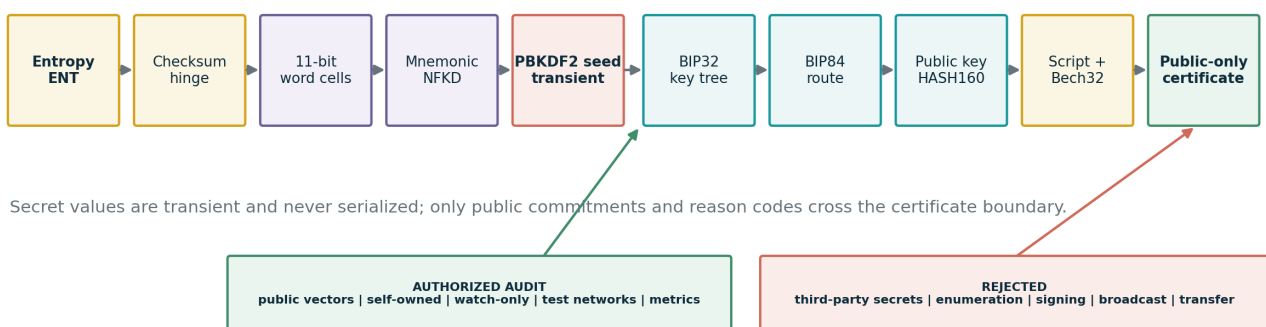


Figure 1: SARA maps the public standards pipeline into a certificate while keeping secret material transient and attack capabilities outside the profile.

2 Referential cryptographic state

2.1 Typed state tuple

The abstract state is

$$q_{\text{SARA}} = (P, E, C, W, M, S, N, D, K, H, Q, A, Z, \mathcal{A}, \mathcal{L}, U),$$

where:

- P is the selected standards/profile identifier;
- E is entropy and C its checksum prefix;
- W is the ordered 11-bit word-cell sequence and M the normalized mnemonic;
- S is the transient 64-byte seed and N a transient HD node;
- D is the derivation path;
- K is a compressed public key and $H = \text{RIPEMD160} \circ \text{SHA256}(K)$;
- Q is the witness program/script relation and A the human-readable address;
- Z is a public certificate;
- $\mathcal{A}$ is authorization state, $\mathcal{L}$ lineage and U uncertainty/status.

2.2 Secret/public partition

FORMAL DEFINITION

Let $\mathcal{S}$ be the secret class and $\mathcal{P}$ the public class:

$$\mathcal{S} = \{E, M, \text{passphrase}, S, k, c, \text{xprv}, \text{WIF}\},$$

$$\mathcal{P} = \{\text{profile}, D, K, H, Q, A, \text{reason codes}, \text{public test vectors}\}.$$

A production certificate is valid only if its serialized field set is disjoint from $\mathcal{S}$. The reference package additionally has no network client and no transaction capability.

The sample certificate uses a universally published test mnemonic, but it still omits mnemonic- and seed-derived fingerprints. The retained four-byte BIP32 parent fingerprint is only a routing hint and is not treated as a collision-free identity.

2.3 Referential definitions

Each definition record contains a stable ID, type, domain, codomain, evaluation phase, dependencies, parameters, invariants, provenance and canonical SHA-256 content hash. References are resolved before evaluation; missing references, duplicate IDs, dependency cycles and hash mismatches are validation failures.

3 BIP39 as an ordered word-cell complex

3.1 Entropy, checksum and cells

For $\text{ENT} \in \{128, 160, 192, 224, 256\}$,

$$\text{CS} = \frac{\text{ENT}}{32}, \quad \text{MS} = \frac{\text{ENT} + \text{CS}}{11}.$$

For entropy bit string e , define

$$b = e \parallel \text{prefix}_{\text{CS}}(\text{SHA256}(e)).$$

The bit string b is partitioned into ordered cells

$$w_i = b[11i : 11(i+1)] \in \{0, \dots, 2047\},$$

and each w_i indexes the committed 2048-word list.

BIP39 as an ordered 1D cell complex

128 entropy bits + 4 checksum bits = 132 bits = 12 cells x 11 bits

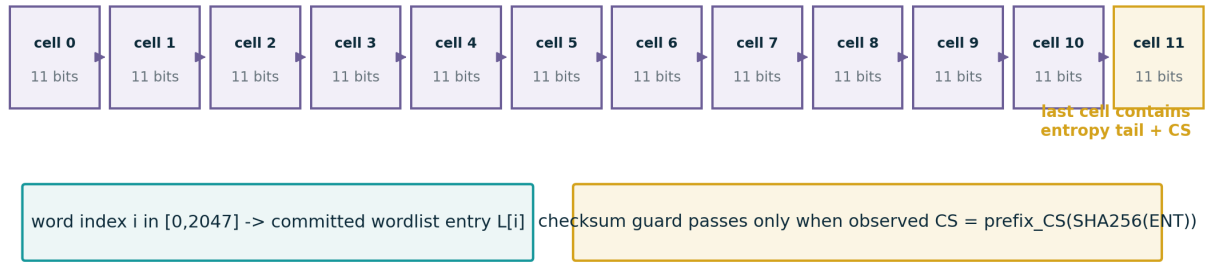


Figure 2: A 12-word mnemonic is represented as a path of twelve labeled 11-bit cells; the final cell contains both entropy and checksum bits.

3.2 Cell-complex definition

Let

$$X_M = (V, E, \lambda), \quad V = \{v_0, \dots, v_m\}, \quad E = \{(v_i, v_{i+1})\}_{i=0}^{m-1},$$

with label map $\lambda(v_i) = w_i$. This is a one-dimensional path complex. Its topology records order and adjacency; it does not add cryptographic strength. The wordlist commitment fixes the meaning of every label by count, uniqueness, order, normalization and digest:

BIP39 English list SHA-256:

2f5eed53a4727b4bf8880d8f3f199efc90e58503646d9ff8eff3a2ed3b24dbda

3.3 Checksum hinge

For a candidate cell path, split the recovered bit string into entropy e' and observed checksum c' . The checksum guard is

$$g_{39}(M) = \begin{cases} 0, & c' = \text{prefix}_{\text{CS}}(\text{SHA256}(e')), \\ 1, & \text{otherwise.} \end{cases}$$

The path is syntactically admitted only when $g_{39} = 0$.

SECURITY / EVIDENCE BOUNDARY

The checksum is an integrity rule, not an ownership test, password test or balance test. A BIP39 mnemonic transports computer-generated entropy; arbitrary memorable sentences are outside this profile. For 12 words, one in sixteen arbitrary 12-index sequences passes the checksum, but the valid space still contains 2^{128} mnemonics.

3.4 Public test vector

The package includes the official public 12-word vector

abandon abandon abandon abandon abandon abandon
abandon abandon abandon abandon abandon about

with indices (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 3) and checksum bits 0011. It is a public conformance vector and must never be used to hold funds.

4 Seed transition and transient-secret boundary

After Unicode NFKD normalization, BIP39 defines

$$S = \text{PBKDF2}_{\text{HMAC-SHA512}}(\text{NFKD}(M), \text{NFKD}(\text{mnemonic}||p), 2048, 64),$$

where p is the optional passphrase. The output S is a 64-byte seed.

FORMAL DEFINITION

The seed transition is an in-memory edge, not a persistent substrate node. The implementation may hold M , p , S , a private scalar and chain code long enough to derive a public test result, but none may cross the certificate serialization boundary. The runtime returns public commitments, path metadata, reason codes and lineage only.

Every passphrase produces a seed; therefore the standard does not provide an intrinsic “wrong passphrase” signal. SARA does not implement passphrase guessing or candidate testing. Self-owned recovery belongs in dedicated audited custody software, not in this reference profile.

5 BIP32 derivation as a typed tree topology

5.1 Extended nodes

A private extended node is

$$n = (k, c, d, f, i),$$

with private scalar k , chain code c , depth d , parent fingerprint f and child number i . The master node is obtained by

$$I = \text{HMAC-SHA512}_{\text{Bitcoin seed}}(S), \quad k = \text{parse}_{256}(I_L), \quad c = I_R.$$

For child index i , the data term is

$$\text{data}(i) = \begin{cases} 0x00 \parallel \text{ser}_{256}(k) \parallel \text{ser}_{32}(i), & i \geq 2^{31}, \\ \text{ser}_P(K) \parallel \text{ser}_{32}(i), & i < 2^{31}, \end{cases}$$

followed by

$$I = \text{HMAC-SHA512}_c(\text{data}(i)), \quad k_i = (\text{parse}_{256}(I_L) + k) \bmod n, \quad c_i = I_R.$$

Invalid zero/out-of-range results are rejected according to the standard.

5.2 Normal and hardened edges

BIP32 derivation topology: normal and hardened edges are different

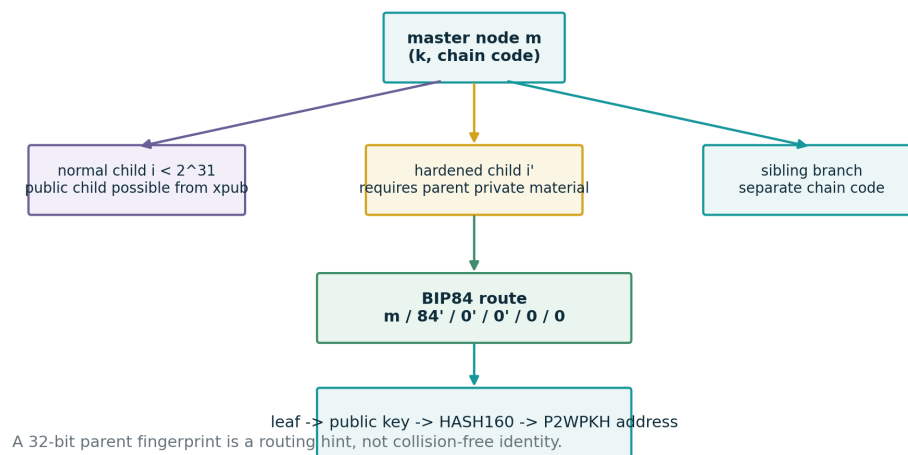


Figure 3: Normal and hardened edges have different derivability and privacy properties; the parent fingerprint is a routing hint, not a collision-free identity.

Normal children use indices $0 \leq i < 2^{31}$ and may be derived from an extended public parent. Hardened children use $i \geq 2^{31}$ and require private parent material. The edge type is therefore part of the topology, not a formatting detail.

5.3 BIP44/BIP84 route word

SARA interprets a path as an ordered edge word. The standard native-SegWit route is

$$D = m/84'/0'/0'/change/index.$$

The reference receiving leaf is

$$m/84'/0'/0'/0'/0/0.$$

The first three edges are hardened; change and address-index edges are normal. This route is a deterministic namespace, not evidence of ownership.

6 Public-key, script and address projection

6.1 P2WPKH relation

For private scalar k , let $K = \text{ser}_P(kG)$ be the compressed public key and

$$h = \text{RIPEMD160} \circ \text{SHA256}(K).$$

A witness-v0 20-byte program has scriptPubKey

$$Q = 0x00\ 0x14 \parallel h.$$

The mainnet human-readable address is the Bech32 encoding of HRP bc, witness version 0 and program h .

6.2 Checksum family guard

For witness version 0, SARA requires Bech32. For witness versions 1–16, it requires Bech32m. The decoder rejects mixed case, invalid characters, checksum failure, invalid witness versions and invalid program lengths. A valid checksum detects many transcription errors; it does not authenticate a person or prove spending authority.

Public address anatomy and exact decode fixture

bc1q7ydrtdn8z62xhslqyqtyt38mm4e2c4h3mxjkug



Valid checksum and P2WPKH classification are public structural facts; they do not reveal a seed or private key.

witness program: f11a35b66716946bc3e0201645c4fbdd72ac56f1

scriptPubKey: 0014f11a35b66716946bc3e0201645c4fbdd72ac56f1

Figure 4: The supplied address is decomposed into public structural fields. No private or signing material is present.

7 Exact public fixtures

7.1 Official BIP84 conformance vector

The public reference path produces:

```

path:      m/84'/0'/0'/0/0
pubkey:    0330d54fd0dd420a6e5f8d3624f5f3482cae350f79d5f0753bf5beef9c2d91af3c
HASH160:   c0cebcd6c3d3ca8c75dc5ec62ebe55330ef910e2
script:     0014c0cebcd6c3d3ca8c75dc5ec62ebe55330ef910e2
address:    bc1qcr8te4kr609gcawutmrza0j4xv80jy8z306fyu
  
```

The certificate contains no mnemonic, passphrase, seed, private scalar, chain code, xprv or WIF field.

7.2 Requester-supplied public address

The decoder returns:

Field	Exact value
Address	bc1q7ydrtdn8z62xhslqyqtyt38mm4e2c4h3mxjkug
HRP	bc
Checksum family	Bech32; checksum valid
Witness version	0
Witness program	f11a35b66716946bc3e0201645c4fbdd72ac56f1
Program length	20 bytes
Address type	P2WPKH
scriptPubKey	0014f11a35b66716946bc3e0201645c4fbdd72ac56f1
Allowed use	Public format/checksum/witness-program/script decode only

SECURITY / EVIDENCE BOUNDARY

These fields establish only that the string is structurally a valid mainnet witness-v0 20-byte address. The package does not establish who controls it, whether it currently holds funds, whether a third-party label is accurate, or whether any testing is authorized by an owner or court.

8 Authorized defensive audit kernel

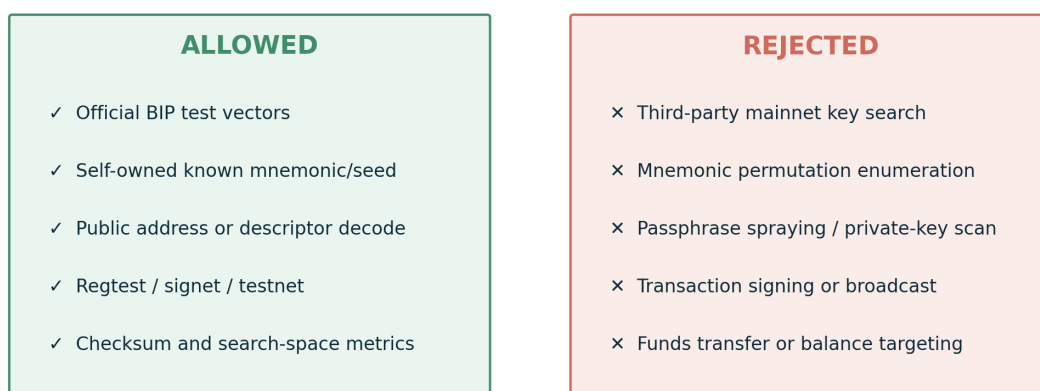
8.1 Authorization predicate

Let s be a requested scope set. The authorization guard is

$$\mathcal{A}(s) = 1 \iff s \subseteq \mathcal{S}_{\text{allowed}} \wedge s \cap \mathcal{S}_{\text{forbidden}} = \emptyset.$$

Allowed scopes include public BIP vectors, public address decode, watch-only consistency, checksum validation, self-owned known-seed derivation, regtest/signet/testnet experiments and non-enumerating metrics. Forbidden scopes include third-party mainnet key search, mnemonic permutations, passphrase spraying, private-key scans, target-address preimage search, signing, broadcast and funds transfer.

Authorized defensive audit boundary



The reference runtime has no network client, candidate generator, signing API or transaction path.

Figure 5: The reference runtime is a conformance and public-structure tool, not a wallet-attack or transaction tool.

8.2 Public-only certificate

A certificate is

$$Z = (P, W_f, D, K, h, Q, A, R, \mathcal{L}),$$

where W_f is a wordlist/checksum fingerprint set, R contains reason codes and all other fields are public commitments or routing metadata. The certificate is issued only when:

1. authorization passes;
2. schema and content hashes pass;
3. checksum and derivation guards pass for the declared test/self-owned profile;
4. address encoding round-trips;
5. secret egress count is zero;
6. network and transaction capabilities are false.

8.3 Pragmatic uses

Within the declared boundary, SARA supports:

- wallet-library conformance against official public vectors;
- self-owned backup consistency checks when the operator already knows the mnemonic/seed;
- watch-only address/script/descriptor audits without spending keys;
- custody-process checks that verify path, network, address type and public commitments;
- testnet, signet or regtest exercises for educational and integration testing;
- public-address parsers, QR/import validators and transcription-error detection;
- search-space and risk communication without candidate generation.

9 Search-space metrics without brute force

9.1 Estimator

For a uniform b -bit space and declared rate $R > 0$, SARA reports

$$N = 2^b, \quad \mathbb{E}[G] = 2^{b-1}, \quad \mathbb{E}[T] = \frac{2^{b-1}}{R}.$$

It does not instantiate a candidate, derive a trial address or compare a candidate to a target.

Declared space	Rate	Expected time	Interpretation
20-bit toy	$10^6/\text{s}$	0.524288 s	Scale illustration only; not Bitcoin compatible.
12-word valid BIP39	$10^{15}/\text{s}$	$5.3914487623 \times 10^{15}$ years	2^{128} valid mnemonics; rate unrealistically favors an attacker.
P2WPKH target preimage	$10^{15}/\text{s}$	$2.3156096112 \times 10^{25}$ years	160-bit target-preimage model; not a collision estimate.
secp256k1 scalar	$10^{15}/\text{s}$	$1.8346149460 \times 10^{54}$ years	Uniform 256-bit scalar model.

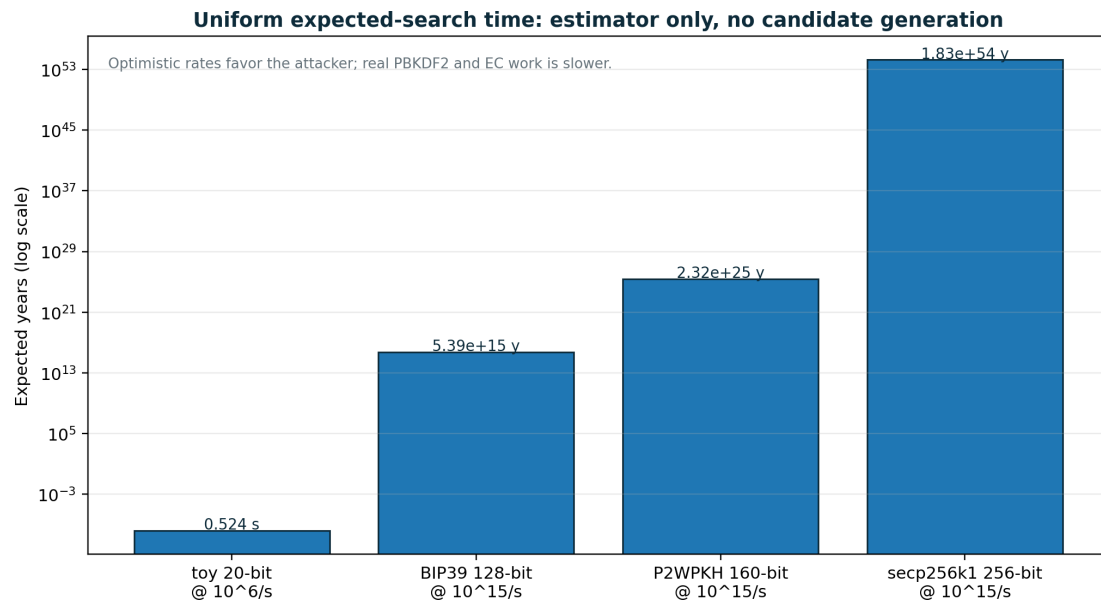


Figure 6: The logarithmic chart shows the expected-time gap between a toy space and cryptographic spaces. The package contains the estimator only.

DERIVATION / INVARIANT

For twelve words, the raw index sequence has 132 bits. The checksum fixes four of them as a function of the other 128, so exactly 2^{128} valid encodings remain. Checksum filtering reduces arbitrary 12-word sequences by a factor of 16; it does not turn the valid space into a tractable search.

SECURITY / EVIDENCE BOUNDARY

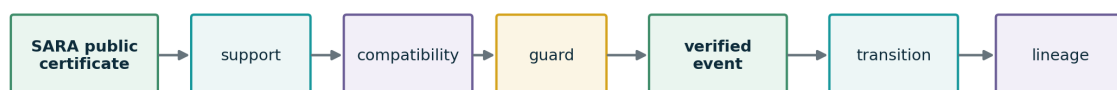
The displayed rates ignore PBKDF2 cost, elliptic-curve work, memory, I/O and coordination and therefore favor the hypothetical searcher. They are feasibility lower bounds, not performance claims. The reference implementation has no candidate generator. Any appearance of a real-wallet target immediately fails the authorization policy.

10 Canonical UGTS event handoff

SARA certification is upstream of the established event sequence:

certificate → support → compatibility → guard → verified event → transition → lineage.

SARA certification returns to the canonical UGTS event calculus



A valid event requires authorized scope, valid schema/checksum/derivation and zero secret egress.

Figure 7: Cryptographic certification does not replace the UGTS event calculus; it supplies a typed public relation and authorization result.

A practical mapping is:

UGTS stage	SARA meaning
Support	Requested audit scope is local, declared and authorized.
Compatibility	Network, address family, derivation profile, schema and wordlist commitment agree.
Guard	Checksum, public vector, round-trip, secret-egress and capability predicates pass.
Verified event	A public-only certificate is issued with reason codes and residual/status fields.
Transition	Public state or watch-only metadata may be updated; secret state is never appended.
Lineage	Profile version, definition hashes, public inputs, test-vector IDs and audit decisions are recorded.

11 Reference implementation

11.1 Modules

Path	Responsibility
src/ugts36/sara363.py	BIP39 validation, BIP32/BIP84 public test derivation, Bech32/Bech32m codecs, public address decode, authorization and metrics.
src/ugts36/sara_runtime.py	Six-step traceable, non-networked public certificate runtime.
spec/SARA_3_6_3_FORMAL_DEFINITION.md	Concise normative definition.
spec/ugts_kc_3_6_3_sara.schema.json	JSON Schema for the literal substrate.
examples/ugts_kc_3_6_3_sara_example.json	Nineteen content-addressed definitions and pipeline references.
data/bip39_english.txt	Official ordered 2048-word public list fixture.
data/sara363_public_address_fixture.json	Public decode result for the supplied address.
data/sara363_search_space_metrics.*	Estimator outputs; no candidate material.
tests/test_sara363.py	Forty-two SARA tests.

11.2 Bounded API

The public reference functions include wordlist commitment, entropy-to-mnemonic conversion for known entropy, mnemonic checksum validation, known-input seed transition, BIP84 public certificate derivation, Bech32/Bech32m encode/decode, public-address decode, authorization and work estimation. The runtime deliberately exposes no search, signing, PSBT, network, balance or broadcast API.

```
PYTHONPATH=src python examples/demo_sara363.py
PYTHONPATH=src python -m unittest discover -s tests -v
```

11.3 Implementation boundary

The elliptic-curve code is a dependency-free oracle for test vectors, not a production wallet library. Production systems require hardened, reviewed cryptographic libraries, secure entropy, side-channel protection, zeroization, isolated signing and audited custody. Passing this report's tests establishes internal conformance only.

12 Validation and exact package metrics

EXACT METRICS

SARA 3.6.3 tests	42 passed
Retained 3.6/3.6.1/3.6.2 tests	153 passed
Combined tests	195 passed
SARA operator records	45
Content-addressed definitions	19
Claims-ledger entries	20
JSON Schema validation errors	0
Invalid definition hashes	0
Reference dependency cycles	0
Official BIP84 public vector matches	3
Supplied-address Bech32 round-trip	pass
Public certificate secret fields	0
Runtime network capability	false
Runtime transaction capability	false
Reference certificate valid	true

The full test suite covers wordlist count and digest, entropy/checksum segmentation, known BIP39 seed output, BIP32 path parsing and child derivation, three BIP84 public vectors, Bech32 and Bech32m vectors, mixed-case and checksum rejection, witness-program length rules, authorization rejection, estimator bounds, secret-free serialization, JSON Schema, content hashes, dependency order and requester attribution.

13 Claims ledger and kill criteria

13.1 Kill criteria

The profile is stopped or simplified when any of the following occurs:

- scope cannot be proven public, self-owned or test-network authorized;
- a secret appears in a serialized definition, trace, log or certificate;
- a new function adds candidate enumeration, target comparison, signing, broadcast or funds transfer;
- a checksum result is presented as ownership or authentication;
- a public address label is presented as legal proof;
- a production claim relies only on dependency-free oracle code or public vectors;
- path, network, witness version, checksum family or program length is implicit rather than typed;
- performance rhetoric is substituted for measured error, rate and authorization boundaries.

13.2 Complete SARA claims ledger

ID	Claim	Disposition	Technical note
SARA-C001	A BIP39 mnemonic is a human-created sentence or brain-wallet.	REJECT	BIP39 transports computer-generated entropy and explicitly warns against user-created sentences.
SARA-C002	The checksum adds independent secret entropy.	CORRECT	The checksum constrains valid encodings and catches some errors; valid mnemonic count remains 2^{ENT} .
SARA-C003	A valid mnemonic checksum proves the owner or wallet balance.	REJECT	Checksum validity is syntax/integrity only.
SARA-C004	The PBKDF2 step makes weak user passphrases unguessable.	BOUND	BIP39 fixes 2048 iterations; passphrase quality remains security-critical.
SARA-C005	A public Bitcoin address can be inverted to its private key.	REJECT	The address exposes a script commitment and checksum, not spend authority.
SARA-C006	Any hash collision can spend a specific P2WPKH output.	REJECT	A target preimage matching the witness program and a valid signature path are required; generic collision rhetoric is not sufficient.
SARA-C007	A Bech32 checksum authenticates the wallet owner.	REJECT	It detects transcription errors; it is not authentication.
SARA-C008	An address label on a block explorer proves legal ownership.	REJECT	Labels are external metadata and require independent provenance.
SARA-C009	A 32-bit BIP32 parent fingerprint is a collision-free identity.	REJECT	It is a short identifier and implementations must tolerate collisions.
SARA-C010	An extended public key is harmless public information.	BOUND	It cannot sign by itself but can reveal address families, balances and privacy-sensitive structure.
SARA-C011	Hardened and normal derivation edges are interchangeable.	REJECT	Hardened children cannot be derived from an extended public parent.
SARA-C012	A Bitcoin address is the same object as a key or wallet.	REJECT	Address, script, public key, extended node, seed and mnemonic are distinct typed objects.
SARA-C013	The supplied public address authorizes private-key testing because it is public.	REJECT	Public visibility does not confer authorization to access or spend funds.
SARA-C014	Brute force against a uniformly random 12-word mnemonic is a practical test plan.	REJECT	The valid keyspace is 2^{128} and the reference metric is astronomically large even at unrealistic rates.
SARA-C015	Address-targeted key search belongs in the reference package.	REJECT	The package provides decoding and feasibility metrics only, not search or candidate generation.
SARA-C016	Mainnet and test networks are equivalent audit environments.	REJECT	Training and destructive experiments belong on regtest, signet, testnet or self-owned fixtures.
SARA-C017	Seed or mnemonic may be stored in the UGTS lineage log.	REJECT	Secrets are transient and prohibited from definitions, traces and certificates.
SARA-C018	The same mnemonic words translated into another language preserve the seed.	REJECT	BIP39 seed derivation hashes the normalized sentence; translation changes the seed.
SARA-C019	One bit can represent the full wallet state.	REJECT	Checksums, keys, paths, scripts, policy, uncertainty and lineage remain separate.
SARA-C020	Passing public test vectors proves production wallet security.	REJECT	It establishes reference conformance only; production security needs hardened libraries, secure entropy and key custody.

14 Final formal definition

FORMAL DECISION

UGTS-KC 3.6.3 SARA is a content-addressed, typed, public-certificate layer over deterministic-wallet standards. It maps BIP39 words into an ordered one-dimensional cell complex; validates the checksum as a guard; treats the seed and private HD nodes as transient; represents BIP32 derivation as a labeled tree with distinct normal and hardened edges; projects only to public keys, witness programs, scripts and addresses; decodes the supplied address as a public fixture; evaluates search-space size without enumeration; rejects unauthorized secret/funds-access scopes; and hands a valid public certificate to the canonical UGTS support-compatibility-guard-event-transition-lineage sequence.

The title “SARA” is an engineering-derived abbreviation for *Seed-Address Referential Algebra*. It is not a Bitcoin standard name and does not imply endorsement by Bitcoin contributors, a wallet provider or a public authority.

15 Complete SARA 3.6.3 operator catalog

ID	Domain	Name	Formal definition	Disposition	Validation
sara363.profile.v1	Profile	Seed-address referential algebra	Typed BIP39/BIP32/BIP84/Bech32 profile with public-only certificates and an authorization gate.	ENGINEERING-DERIVED	Implemented
sara363.governance.evidence-boundary	Governance	Source and standards boundary	Separate BIP-defined transformations, engineering glue, public fixtures, metrics and rejected attack scopes.	RETAIN	Enforced
sara363.security.secret-class	Security	Secret/public classification	Entropy, mnemonic, passphrase, seed, private scalar and chain code are secret; addresses, scripts and test vectors are public.	ENGINEERING-DERIVED	Implemented
sara363.wordlist.commitment	Encoding	2048-word list commitment	Bind the selected BIP39 wordlist by count, order, uniqueness, normalization and SHA-256 digest.	RETAIN	Exact
sara363.entropy.input	Encoding	Computer-generated entropy input	Admit only ENT in {128,160,192,224,256}; do not treat arbitrary user sentences as entropy.	RETAIN	Exact
sara363.checksum.prefix	Encoding	BIP39 checksum prefix	CS=ENT/32 and checksum bits are the first CS bits of SHA-256(entropy).	RETAIN	Exact
sara363.bits.concat	Encoding	Entropy-checksum concatenation	Append checksum bits to entropy before word segmentation.	RETAIN	Exact
sara363.bits.segment11	Encoding	Eleven-bit segmentation	Split ENT+CS into ordered 11-bit cells, each in [0,2047].	RETAIN	Exact
sara363.words.lookup	Encoding	Word-index lookup	Map each 11-bit cell to one word in the committed list.	RETAIN	Exact
sara363.words.cell-complex	Topology	Ordered word-cell complex	Represent the mnemonic as a 1D path of labeled 11-bit cells whose adjacency preserves word order.	TRANSLATE	Specified
sara363.mnemonic.checksum-hinge	Event calculus	Mnemonic checksum hinge	A mnemonic is admitted only when observed checksum cells equal the entropy-hash prefix.	TRANSLATE	Implemented
sara363.text.nfkd	Encoding	Unicode NFKD normalization	Normalize mnemonic and passphrase as required before seed derivation.	RETAIN	Implemented
sara363.seed.pbkdf2	Cryptography	Mnemonic-to-seed transition	PBKDF2-HMAC-SHA512 with 2048 iterations, 64-byte output and salt 'mnemonic'+passphrase.	RETAIN	Implemented
sara363.seed.no-serialization	Security	Seed nonserialization	Seed bytes are transient and excluded from substrate definitions, traces, certificates and package data.	ENGINEERING-DERIVED	Enforced
sara363.bip32.master	Cryptography	BIP32 master node	HMAC-SHA512 with key 'Bitcoin seed' creates a private scalar and 32-byte chain code.	RETAIN	Implemented
sara363.bip32.node	Topology	Extended-key tree node	Node state is key material plus chain code, depth, parent fingerprint and child number.	RETAIN	Implemented
sara363.bip32.normal-edge	Topology	Normal derivation edge	Indices below 2^{31} admit private and public child derivation under BIP32 rules.	RETAIN	Private path implemented
sara363.bip32.hardened-edge	Topology	Hardened derivation edge	Indices at or above 2^{31} require parent private material and block public-parent derivation.	RETAIN	Private path implemented
sara363.path.parser	Encoding	Derivation-path parser	Parse m/... with explicit hardened markers and uint32 bounds.	ENGINEERING-DERIVED	Implemented
sara363.path.bip44	Information tecture	archi- Five-level account hierarchy	Purpose/coin/account/change/index are distinct typed path levels.	RETAIN	Specified
sara363.path.bip84	Information tecture	archi- Native P2WPKH derivation policy	Use m/84'/coin'/account'/change/index for BIP84 accounts.	RETAIN	Implemented
sara363.ec.secp256k1	Cryptography	secp256k1 public projection	Map a valid scalar to a curve point without exposing a signing API.	RETAIN	Reference implementation
sara363.pubkey.compressed	Encoding	Compressed public key	Serialize a curve point as parity prefix plus 32-byte x coordinate.	RETAIN	Implemented
sara363.hash.hash160	Cryptography	HASH160 commitment	RIPEMD160(SHA256(compressed public key)) yields the 20-byte P2WPKH witness program.	RETAIN	Implemented
sara363.script.p2wpkh	Relation	P2WPKH script relation	Witness v0 with a 20-byte key hash maps to scriptPubKey 0x0014{program}.	RETAIN	Implemented
sara363.address.convertbits	Encoding	Eight-to-five-bit regrouping	Convert witness-program bytes into 5-bit Bech32 data groups with canonical padding.	RETAIN	Implemented
sara363.address.hrp	Encoding	Human-readable prefix	Bind the network namespace through HRP such as bc or tb.	RETAIN	Implemented
sara363.address.bech32	Encoding	Bech32 checksum shell	Witness version 0 addresses use the BIP173 polymod constant.	RETAIN	Implemented
sara363.address.bech32m	Encoding	Bech32m checksum shell	Witness versions 1-16 use the BIP350 polymod constant.	RETAIN	Implemented
sara363.address.decode	Query	Public address decoder	Validate case, HRP, checksum family, witness version, program length and script projection.	ENGINEERING-DERIVED	Implemented
sara363.address.classify	Query	Witness-program classifier	Classify v0/20 as P2WPKH, v0/32 as P2WSH and v1/32 as a Taproot-compatible form.	ENGINEERING-DERIVED	Implemented
sara363.address.noninverse	Security	Non-reversibility boundary	Address decoding exposes a script commitment, not a private key, seed or spend authority.	CORRECTED	Enforced

ID	Domain	Name	Formal definition	Disposition	Validation
sara363.fixture.public-address	Fixture	User-supplied public address fixture	Use the supplied bc1q address only for format, checksum, witness-program and scriptPubKey validation.	BOUNDED	Verified
sara363.audit.self-owned	Audit	Self-owned known-seed audit	Permit deterministic backup verification only when the operator already possesses the mnemonic or seed.	BOUNDED	Policy
sara363.audit.watch-only	Audit	Watch-only public audit	Permit address/descriptor parsing and public derivation evidence without spend authority.	BOUNDED	Policy
sara363.audit.authorization-gate	Governance	Authorization gate	Every audit declares a scope and is rejected if any operation targets private keys, third-party secrets or funds.	ENGINEERING-DERIVED	Implemented
sara363.audit.forbidden-search	Governance	Forbidden-target rejection	Reject mnemonic enumeration, passphrase spraying, private-key scanning and address-targeted preimage search.	REJECT	Enforced
sara363.audit.no-network	Security	No-network reference runtime	The reference implementation performs no blockchain queries, balance lookups or remote calls.	ENGINEERING-DERIVED	Enforced
sara363.audit.no-signing	Security	No-signing reference runtime	The reference implementation exposes no transaction signing, PSBT mutation, broadcast or funds-transfer path.	ENGINEERING-DERIVED	Enforced
sara363.metric.uniform-search	Metric	Uniform search-space estimator	Report 2^b candidates, $2^{(b-1)}$ expected guesses and time at a declared optimistic rate; do not enumerate.	ENGINEERING-DERIVED	Implemented
sara363.metric.bip39-valid-space	Metric	Valid mnemonic keyspace	A 12-word BIP39 space contains 2^{128} valid mnemonics; checksum filtering does not reduce the encoded entropy below ENT.	CORRECTED	Exact
sara363.metric.toy-space	Metric	Reduced synthetic metric	Use a 20-bit toy space only to explain scale; it is explicitly not a Bitcoin wallet or key search.	BOUNDED	Measured
sara363.certificate.public-only	Certificate	Public-only derivation certificate	Return path, parent routing fingerprint, compressed public key, HASH160, script and address while excluding mnemonic-, seed- and private-key-derived commitments.	ENGINEERING-DERIVED	Implemented
sara363.lineage.audit-record	Persistence	Audit lineage record	Log standards profile, public fixture digest, reason codes and test-vector result, never mnemonic/seed/private keys.	ENGINEERING-DERIVED	Specified
sara363.handoff.ugts	Architecture	Canonical UGTS handoff	After cryptographic certification, return to support, compatibility, guard, event, transition and lineage.	RETAIN	Specified

16 Package layout and reproduction

README.md / NOTICE.md / CHANGELOG.md / VERSION
report/ PDF, editable TeX, figures, preflight reports
spec/ formal definition, 45 operators, 20 claims, JSON Schema
src/ugts36/ SARA runtime plus retained 3.6/3.6.1/3.6.2 modules
tests/ 195-test combined suite
examples/ literal substrate, public certificate and trace
data/ BIP39 wordlist, public address fixture, metrics
sources/ standards register and privacy-safe distillation notes
checksums/ SHA-256 manifest and package verification

No raw private wallet material, raw uploaded source PDFs, font files, network credentials or transaction artifacts are included.

Normative references

- BIP 39, *Mnemonic code for generating deterministic keys*: <https://github.com/bitcoin/bips/blob/master/bip-0039.mediawiki>
- BIP 32, *Hierarchical Deterministic Wallets*: <https://github.com/bitcoin/bips/blob/master/bip-0032.mediawiki>
- BIP 44, *Multi-account hierarchy*: <https://github.com/bitcoin/bips/blob/master/bip-0044.mediawiki>
- BIP 84, *Derivation scheme for P2WPKH accounts*: <https://github.com/bitcoin/bips/blob/master/bip-0084.mediawiki>
- BIP 141, *Segregated Witness*: <https://github.com/bitcoin/bips/blob/master/bip-0141.mediawiki>
- BIP 173, *Base32 address format for native witness outputs*: <https://github.com/bitcoin/bips/blob/master/bip-0173.mediawiki>
- BIP 350, *Bech32m format for v1+ witness addresses*: <https://github.com/bitcoin/bips/blob/master/bip-0350.mediawiki>
- UGTS-KC 3.6.2 SCLP, inherited referential geometry/topology, packed state and canonical event handoff.